

Enhancing Face Recognition Attendance System Utilizing Real-Time Face Tracking

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Abstract

Manual roll calls and existing biometric attendance systems, which capture attendance only at the start or end of class, are prone to inaccuracies such as proxy attendance and fail to monitor students' presence throughout sessions. This study proposes an enhanced face recognition attendance system utilizing real-time face tracking to ensure accuracy and reliability in attendance tracking. The system captures facial data using OpenCV for detection and a CNN-based library for recognition, logging attendance at 30-minute intervals. A minimum presence of 80% of the session must be marked present. Attendance records are synchronized in real-time using Firebase, and insights are generated using Plotly for visual analytics. The system achieves a recognition accuracy of 94% under optimal conditions and demonstrates robustness under varying environments. Comparative analysis with existing algorithms highlights its improved scalability and usability, significantly advancing over traditional methods.

Keywords

Real-time Face Tracking, Attendance System, Convolutional Neural Networks, Firebase, Visual Analytics

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